

Workshop on Advanced Sampling Methodologies

Google Earth Engine Code Editor Tutorial

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1. Open the Code Editor

- Go to <https://code.earthengine.google.com/>.
 - Make sure you're signed in with a Google account that has Earth Engine access.
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2. Load a Shapefile

There are two ways:

Option A: Upload a shapefile

1. In the **Assets** tab (left panel), click **NEW → Table upload**.
2. Upload your **.shp** (plus **.shx**, **.dbf**, etc. in a zipped folder).
3. Once uploaded, it appears under your assets.

Option B: Use an existing dataset

```
// Example: Use a country boundary from FAO GAUL dataset
var region = ee.FeatureCollection("FAO/GAUL/2015/level0")
               .filter(ee.Filter.eq('ADM0_NAME', 'Kenya'));
Map.centerObject(region, 6);
```

3. Load an image and subset to shapefile

Example with **Sentinel-2 surface reflectance**:

```
// Load Sentinel-2 surface reflectance
var image = ee.ImageCollection("COPERNICUS/S2_SR")
               .filterDate('2021-01-01', '2021-12-31')
               .filterBounds(region)
               .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 10))
               .median();

// Clip the image to shapefile boundaries
var clipped = image.clip(region);
```

4. Explore available bands

Check which bands are included:

```
print('Sentinel-2 bands:', image.bandNames());
```

Example band names: B2 (blue), B3 (green), B4 (red), B8 (NIR).

5. Visualize the image

```
// True color (RGB) visualization
var visParams = {
  bands: ['B4', 'B3', 'B2'],
  min: 0,
  max: 3000,
};

Map.addLayer(clipped, visParams, 'Sentinel-2 RGB');

// False color (NIR, Red, Green)
var falseColor = {
  bands: ['B8', 'B4', 'B3'],
  min: 0,
  max: 3000,
};
Map.addLayer(clipped, falseColor, 'False Color');
```

6. Export the image

Export to **Google Drive**:

```
Export.image.toDrive({
  image: clipped,
  description: 'Sentinel2_Kenya_2021',
  folder: 'EarthEngineExports',
  scale: 10, // Sentinel-2 resolution
  region: region.geometry(),
  maxPixels: 1e13
});
```

Export to **Asset** (for later use in GEE):

```
Export.image.toAsset({
  image: clipped,
  description: 'Sentinel2_Kenya_Asset',
```

```
assetId: 'users/your_username/Sentinel2_Kenya',  
scale: 10,  
region: region.geometry(),  
maxPixels: 1e13  
});
```

Summary

- Upload/load a shapefile as a `FeatureCollection`.
 - Filter an image collection and clip it to your shapefile.
 - Print and inspect band names.
 - Visualize with `Map.addLayer()` using different bands.
 - Export results to Drive or Assets.
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